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Microbial Activity and Ruminal Methanogenesis as Affected by Plant Secondary Metabolites in Different Plant Extracts

S.K.Sirohi, Neha Pandey, Navneet Goel, B. Singh, Madhu Mohini, Poonam Pandey and P.P Chaudhry

Abstract—A study was conducted to evaluate total sugar, total protein, total tannin and total saponin content of aqueous, methanol and acetone extracts of different twenty one herbal plants / plant parts. Further, in vitro dry matter digestibility (IVDMD), methane reduction and antimicrobial potential of plant extracts on the Gram positive and Gram negative bacterial culture were also evaluated. The plants / plant parts taken for the study were satavari-roots (*Asparagus racemosus*), pudina-leaves (*Mentha arvensis*), alstonia-leaves (*Alstonia scholaris*), aloe vera-leaves (*Aloe barbadensis*), gokhru-seeds (*Tribulus terrestris*), yam-root (*Dioscorea bulbifera*), garlic-roots (*Allium sativum*), coriander-leaves (*Coriandrum sativum*), ginger-roots (*Zingiber officinale*), onion-root (*Allium cepa*), neem-leaves (*Azadirachta indica*), ashwagandha-leaves (*Withania somnifera*), beleric-leaves (*Terminalia belerica*), eucalyptus-leaves (*Eucalyptus globulus*), guava-leaves (*Psidium guajava*), arjuna-leaves (*Terminalia arjuna*), harad-leaves (*Terminalia chebula*), shikakai-pods (*Acacia concinna*), reetha-fruit (*Sapindus mukorossi*), ashok-leaves (*Saraca asoca*) and jamun-leaves (*Syzygium cumini*). Different type of extracts were prepared in methanol, acetone and water. These extracts were also tested for total protein, total sugar, total tannin, total saponin and methane production. For estimating the antimicrobial potential of these extracts subjected to four bacterial cultures *Streptococcus*, *Staphylococcus* (gram positive), *Escherichia coli* and *Enterobacter* (gram negative). Methanol, acetone and water were used as control. It was concluded from results that both acetone and methanol extracts of *E. globulus* and aqueous extract *S. mukorossi* and *E. globulus* were the best inhibitor of methane production under in vitro environment. The *Enterobacter* was the most sensitive bacteria which showed the zone against 70% of the extracts tested. The acetone extract was found to be more effective than methanol or aqueous extracts in inhibiting bacterial growth. Among all the extracts tested Jamun (*Syzygium cumini*) gave the maximum zone of inhibition for both Gram positive and Gram negative cultures.

Keywords—Plant extracts, Plant secondary metabolites, Methane, Antimicrobial activity

I. INTRODUCTION

RUMEN is a diverse and unique microbial ecosystem composed of bacteria, protozoa and fungi. In rumen hydrogen is produced during the anaerobic fermentation of nutrients. This hydrogen can be used during the synthesis of volatile fatty acids (VFAs) and microbial protein synthesis.

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The excess hydrogen from NADH is eliminated primarily by the formation of methane produced by methanogens. Methane production in the ruminants is an energetically (2-15% of ingested GE) wasteful process, since the portion of animal's feed, which is converted to methane is eructed as gas. Furthermore, emission of methane in the environment contributes to global warming by trapping outgoing terrestrial infrared radiation. The concentration of Green House Gases in the troposphere has implicated in the consistent increase in the atmospheric pressure and global warming over the last few decade [21]. The rising concentration of methane is correlated with increasing world's population and currently about 70% of the methane production arises from anthropogenic sources and the remainder from natural sources [5]. However the antibiotics as feed additives in beef cattle and dairy cows has been banned in the European Union due to the risk of antibiotics residues in animal products (e.g.: milk and meat) and its subsequent effects on human health. For this reason, attention has recently shifted towards use of natural antimicrobials or Plant Secondary Metabolites (PSM) as a safe means of ruminal fermentation modulators.

Recent studies have shown that plant secondary metabolites (PSM) at lower concentrations could be used to manipulate rumen fermentation favorably. At appropriate dose, saponins or saponins containing plants have been shown to suppress protozoal population, increase bacteria and fungi population, propionate production, partitioning factor, yield and efficiency of microbial protein synthesis and decrease methanogenesis hence improve performance in ruminants [2], [20], [17]. Tannins especially condensed tannins (CT) also decrease methane production and increase efficiency of microbial protein synthesis as reported by [20]. Plant extracts rich in flavonoids increase degradation of cell wall constituents, yield and efficiency of microbial protein synthesis [26]. Therefore, the present experiment was planned to see the effect of different plant extracts on methane and antimicrobial activity under in vitro conditions.

II. MATERIALS AND METHODS

A. Preparation of Plant Extracts

Water, 50% aqueous methanol and 50% aqueous acetone were used as a solvent to prepare plant extracts. The plant part obtained were cut or crushed into small pieces and oven dried low temperature at 30-50°C. The plant material was then ground to pass through 1mm screen. A known quantity of finely ground sample was weighed into 250ml conical flask.

The respective solvents were added (1:10 dried plant to solvent) and the flask was tightly sealed and kept in a shaker at 250C and 120 rpm for 24 hour. After shaking the content of the flask, it is squeezed through four layers of muslin cloth. The filtrate obtained from aqueous extract was centrifuged at 3500 rpm for 20 minutes and then filtered through Whatman No 1 filter paper, while aqueous methanol and aqueous acetone extracts directly filtered using Whatman No 1 filter paper.

B. Estimation of Total Protein

Total protein in the plant extract was determined using calorimetric method described by [31]. Plant extract (0.4ml) was mixed with 4 ml of copper sulphate solution and incubated at room temperature for 10 minutes. Then, 4 ml of phenol reagent was added and allowed to react for 30 minutes. The absorbance was measured at 600 nm against reagent blank. Bovine Serum Albumin (100mg/ml) was used as a standard and then 15, 30, 60, 90, 120, and 150 µg were taken from the standard solution and these readings were used for standard curves.

C. Estimation of Total Sugars

Total sugar in the plant extract was determined using calorimetric method described by [30]. Plant extract (1 ml) was mixed with 1 ml of 2% phenol and 5 ml of concentrated sulphuric acid, allowed to react for 30 minutes and the absorbance was measured at 430 nm against reagent blank. For total sugar estimation 0.05% (50mg/100ml) glucose was used as a standard and then 0.25, 0.50, 0.75 and 1.0 mg were taken from the standard solution and readings were used for drawing standard curves.

D. Estimation of Tannin

Total tannins were determined by method described by [19]. PVPP (100 mg) is taken in a test tube and 1ml of distilled water along with 1ml of plant extract is added to it. It is then centrifuged at 3000 rpm for 10 minutes and the supernatant is collected. This supernatant has only simple phenolics other than tannins, so total tannins measured by subtracting simple phenolics (supernatant fractions) from the total phenolics

E. Estimation of Saponin

Saponin in the plant extract was determined using calorimetric method described by [8] after minor modifications. Saponin prepared from *Asparagus adscendens* root were used as standard. Saponin extract (0.25 ml) was added to a tube containing 1.75 ml ethyl acetate. Then 1.0 ml reagent A (0.5 ml anisaldehyde + 99.5 ml ethyl acetate) and 1.0 ml reagent B (50 ml concentrated H₂SO₄ + 50 ml ethyl acetate) was added and mixed thoroughly and kept in water bath maintained at 600C for 20 minutes. After cooling to room temperature, the absorbance was measured at 430 nm against reagent blank. The amount of steroidal saponin in the extract was calculated as *A. adscendens* saponin equivalent based on the standard curve *A. adscendens* saponins.

F. Estimation of Methane Production

For estimating the methane production incubations were carried out in 100 ml calibrated plastic syringes as described

by [26]. The syringes were incubated in water bath at 39 ± 0.50C [28]. The substrate was weighed on a plastic boat with removable stem and was placed into the bottom of the plastic syringe without sticking to the sides of the syringes. The piston was lubricated with petroleum jelly and pushed inside the plastic syringe. The plant extracts were injected about 1 hour before incubation. The syringes were kept in an incubator at 390C until incubation.

The medium mixture solution was prepared by mixing 500ml distilled water, 0.125 ml micro mineral solution, 250 ml buffer solution and 250ml macro mineral solution, 1.25 ml resazurine solution, and 50 ml reducing solution (prepared fresh and added prior to incubation). The medium mixture solution was pre warmed to 390 C and bubbled with CO₂ just before collection of rumen liquor. Rumen liquor was collected from fistulated male cattle maintained on standard diet (60 part roughage and 40 part concentrate) before morning feeding into a pre warmed thermo flask and brought to laboratory. The rumen liquor was filtered through four layers of muslin cloth. Once the medium mixture solution becomes colorless, the required amount of filtered rumen liquor was added. The proportion of medium mixture solution to rumen liquor was 2:1. Just after mixing the medium and rumen liquor, 30 ml of incubation medium was injected to the syringes using auto dispenser. The syringes were shaken gently and residual air or air bubbles if any was removed and outlet was closed. The level of piston was recorded and syringes were placed in water bath (39± 0.50C). The syringes were shaken every one hour from the start of the incubation up to 10 hour of incubation. These trials were conducted along with respective blank and control in triplicate.

After 24 hour of incubation, volume of gas was withdrawn from the tip of the incubation syringe using Hamilton gas tight syringe and analyzed for methane with the help of gas chromatograph (Nucon 5700, India) fitted with stainless steel column packed with Porapak- Q (length 6'; o.d. 1/8" i.d. 2mm; mesh range 80-100) and Flame ionizing detector (FID). The temperature of injection port, column and detector was 40, 50 and 500C respectively. Volume of gas taken for injecting was 200µl. The flow rate of carrier gas (nitrogen) through the column was 30ml/min and H₂ was 30ml/min and air was 300 ml/min. The standard gas for methane estimation (Spantech calibration gas, Surrey, England) composed of 50% methane and 50% CO₂. The peak of methane gas was identified on the basis of retention time of standard methane gas and the response factor obtained was used to calculate methane percentage in the gas sample. The methane produced from substrate during 24 hour incubation was corrected for the blank values. The volume of methane produced was calculated as follows,

Methane production (ml) = Total gas produced (ml) x % methane in the sample.

G. In vitro Dry Matter Digestibility (IVDMD)

To estimate true DM digestibility of each syringe containing residues after incubation was estimated as per [32]. 200mg of substrate was taken in triplicates in glass syringes (100ml) and incubated for 24 hour duration. After recording the total gas volume, entire syringe content was drained through the spike into 500ml spoutless beakers. The syringe

TABLE I
PLANT SECONDARY METABOLITES AND OTHER IENTS IN DIFFERENT PLANT EXTRACTS (% DM BASIS)

S. No.	Local/ common name	Botanical name	Acetone extract				Methanol extract				Aqueous extract			
			Total Protein	Total Sugar	Total Tannins	Total Saponins	Total Protein	Total Sugar	Total Tannins	Total Saponins	Total Protein	Total Sugar	Total Tannin	Total Saponins
1	Garlic	<i>Allium sativum</i>	1.52	15.38	10.30	4.88	1.38	28.00	7.97	20.94	1.15	22.09	3.98	5.40
2	Yam	<i>Dioscorea bulbifera</i>	0.02	5.75	0.17	0.14	0.02	6.47	2.40	1.02	0.02	1.93	2.47	0.14
3	Ashwgandha	<i>withania somnifera</i>	0.03	5.79	1.68	0.51	0.03	5.26	5.51	2.98	0.02	2.35	1.76	0.19
4	Arjun	<i>Terminalia arjuna</i>	0.04	2.48	0.79	0.43	0.06	2.73	0.14	2.36	0.04	0.87	0.84	0.49
5	Guava	<i>Psidium guajava</i>	0.27	30.26	0.18	6.10	0.23	20.18	22.28	14.66	0.19	8.75	1.56	6.53
6	Alstonia	<i>Alstonia scholaris</i>	0.28	23.20	0.66	4.07	0.26	18.33	0.21	29.32	0.19	15.10	1.45	25.07
7	Eucalyptus	<i>Eucalyptus globulus</i>	0.06	8.94	6.81	1.08	0.06	6.66	2.16	5.36	0.05	2.60	1.43	1.96
8	Gokhru	<i>Tribulus terrestris</i>	0.08	19.15	12.68	0.50	0.07	15.43	11.46	4.11	0.05	4.41	2.44	0.17
9	Reetha	<i>Sapindus mukorossi</i>	0.05	9.43	0.14	0.76	0.05	2.71	1.03	1.56	0.04	1.75	0.47	0.66
10	Ginger	<i>Zingiber officinale</i>	0.17	17.71	0.53	0.10	0.15	10.70	8.10	0.36	0.12	2.38	5.57	0.53
11	Behara	<i>Terminalia belerica</i>	0.07	15.76	0.01	2.33	0.05	13.63	0.19	15.32	0.04	3.02	2.87	1.75
12	Shikakai	<i>Acacia concinna</i>	0.08	11.48	0.38	0.87	0.07	10.18	9.75	3.40	0.06	3.56	1.66	1.05
13	Harad	<i>Terminalia chebula</i>	0.09	14.56	2.46	0.99	0.08	12.90	2.60	5.08	0.06	3.11	4.53	1.61
14	Jamun	<i>Syzygium cumini</i>	0.04	6.91	0.13	1.34	0.03	6.79	5.93	4.66	0.03	1.55	1.28	0.58
15	Neem	<i>Azadiracta indica</i>	0.03	3.64	0.00	0.45	0.03	3.82	0.02	3.21	0.02	1.14	0.06	0.45
16	Aloe vera	<i>Aloe barbadensis</i>	0.06	2.28	0.16	0.01	0.05	3.44	1.91	0.19	0.04	0.74	1.49	0.06
17	Satavari	<i>Asparagus racemosus</i>	0.06	8.31	3.71	0.22	0.05	5.98	0.73	1.93	0.05	2.94	0.29	1.01
18	Ashok	<i>Saraca asoca</i>	1.00	6.08	12.00	6.35	1.05	18.25	9.96	4.85	1.27	4.33	9.02	2.89
19	Coriander	<i>Coriandrum sativum</i>	0.22	22.58	ND	6.29	0.23	25.68	ND	3.48	0.26	8.52	6.07	2.01
20	Onion	<i>Allium cepa</i>	0.31	17.86	4.43	10.61	0.32	19.50	ND	7.11	0.43	23.70	ND	4.64
21	Pudina	<i>Menthe orvensis</i>	0.23	21.65	0.48	8.56	0.18	11.93	0.21	3.18	0.24	7.40	ND	2.18

was washed three times with a total of 70ml of neutral detergent solution (NDS) using dispenser to dispense the NDS through the spike of the syringe. Beakers were kept on heaters and refluxing was done at 1000C for 1 hour from the boiling started. After the boiling was over, contents were filtered through the sintered glass crucible (No 1) and kept in hot air oven at 1000C for drying. The IVDMD was the difference between the weight of the dry matter incubated and the DM residue left. The DM degradability was calculated as weight of DM incubated minus weight of DM residue.

IVDMD (%) = (Wt. of dried sample - Wt. of residue / Wt. of Dried sample) x 100

H. Estimation of Antimicrobial Potential of Plant Extracts

The antimicrobial potential of plant extract was determined by [6] method with some modifications. For estimating the antimicrobial potential of plant extracts *Streptococcus*, *Staphylococcus* (gram positive), *Escherichia coli* and *Enterobacter* (gram negative) bacterial cultures were taken. An

18-24 hour old bacterial cultures were mixed with normal saline (0.85% NaCl) to match the concentration of 0.5 Mac Farland standards (1.5 x 10⁸ cells/ml). 1ml of this was mixed with 19 ml of Nutrient agar maintained at 500C. It was then poured in pre-sterilized Petri-dish and allowed to solidify. Pre sterilized filter paper discs with 4 mm diameter were soaked in plant extracts and placed on the solidified Petri-plates at equal distance. Methanol, acetone and water were used as control. These were then incubated at 370C for 24 hours and observed for the zone of inhibition.

III. RESULTS

The results of total protein, total sugar, total tannin and total saponin content of all the acetone, methanol and aqueous extracts of 21 plants were presented in table I. Among the selected acetone extracts, *A. sativum* contained maximum (1.52%) of total protein, but in contrast the minimum protein was found in *D. bulbifera* (0.02%).

TABLE II
METHANE PRODUCTION AND *IN VITRO* DRY MATTER DIGESTIBILITY OF DIFFERENT PLANT EXTRACTS

S.No.	Local/ Common name	Botanical name	Acetone extract				Methanol extract				Aqueous extract			
			CH ₄ ml/gDM	CH ₄ mMol/ gDM	CH ₄ mMol/ gDDM	IVDMD %	CH ₄ ml/g DM	CH ₄ mMol/ gDM	CH ₄ mMol/ gDDM	IVDMD %	CH ₄ ml/gDM	CH ₄ mMol/ gDM	CH ₄ mMol/ gDDM	IVDMD %
1	Garlic	<i>Allium sativum</i>	5.22	0.52	0.272	52.50	31.52	3.13	1.65	52.50	32.06	3.19	2.34	73.50
2	Yam	<i>Dioscorea bulbifera</i>	13.35	1.33	0.607	45.75	50.22	4.99	2.87	57.50	29.85	2.97	2.27	76.50
3	Ashwagandha	<i>withania somnifera</i>	19.02	1.89	1.253	66.25	45.22	4.50	2.66	59.25	26.26	2.61	1.91	73.25
4	Arjun	<i>Terminalia arjuna</i>	8.24	0.82	0.483	59.00	51.95	5.17	2.13	41.25	30.10	2.99	1.94	64.75
5	Guava	<i>Psidium guajava</i>	17.41	1.73	1.077	62.25	56.10	5.58	3.50	62.75	25.02	2.49	1.33	53.50
6	Alstonia	<i>Alstonia scholaris</i>	22.68	2.26	1.257	55.75	67.18	6.68	4.23	63.25	30.04	2.99	2.31	77.25
7	Eucalyptus	<i>Eucalyptus globulus</i>	0.21	0.02	0.010	48.00	22.81	2.27	1.16	51.00	24.66	2.45	1.77	72.00
8	Gokhru	<i>Tribulus terrestris</i>	21.87	2.17	1.370	63.00	56.28	5.60	3.54	63.25	30.71	3.05	2.31	75.50
9	Reetha	<i>Sapindus mukorossi</i>	12.63	1.26	0.791	63.00	73.90	7.35	4.45	60.50	24.18	2.40	1.35	56.25
10	Ginger	<i>Zingiber officinale</i>	29.99	2.98	2.222	74.50	73.53	7.31	5.56	76.00	27.37	2.72	1.98	72.75
11	Behera	<i>Terminalia belerica</i>	9.10	0.90	0.484	53.50	36.64	3.64	1.77	48.50	29.42	2.93	1.76	60.00
12	Shikakai	<i>Acacia concinna</i>	28.15	2.80	1.351	48.25	45.27	4.50	2.61	58.00	37.02	3.68	2.39	65.00
13	Harad	<i>Terminalia chebula</i>	14.68	1.46	0.843	57.75	32.05	3.19	1.21	38.00	25.34	2.52	1.61	64.00
14	Jamun	<i>Syzgium cumini</i>	17.18	1.71	1.046	61.25	39.71	3.95	1.00	25.25	26.29	2.61	2.10	80.25
15	Neem	<i>Azadiracta indica</i>	22.88	2.28	1.314	57.75	35.61	3.54	2.20	62.25	35.10	3.49	2.39	68.50
16	Aloe vera	<i>Aloe barbadanis</i>	31.85	3.17	2.312	73.00	62.18	6.18	3.83	62.00	29.86	2.97	1.99	67.00
17	Satavari	<i>Asparagus racemosus</i>	106.09	10.55	6.989	66.25	124.76	12.41	7.91	63.75	41.39	4.12	2.86	69.50
18	Ashok	<i>Saraca asoca</i>	31.93	3.18	1.96	60.5	31.52	4.14	2.57	62.00	38.97	3.88	2.56	66.50
19	Coriander	<i>Coriandrum sativum</i>	34.45	3.43	2.51	73.25	50.22	2.17	1.41	54.25	45.29	4.50	3.32	73.75
20	Onion	<i>Allium cepa</i>	45.75	4.55	2.86	73.67	67.18	2.87	1.47	54.75	45.31	5.51	3.38	75.00
21	Pudina	<i>Menthe orvensis</i>	33.59	3.34	2.55	64.00	22.81	4.51	2.72	60.25	46.25	4.60	3.26	71.00
22	Control Diet	50R : 50C	37.55	3.73	2.94	63.25	37.85	3.76	2.67	70.50	29.45	2.93	1.96	74.75

In aqueous methanol extract *A. sativum* contained the maximum amount of protein i.e. 1.38% while the lowest (0.02%) was found in *D. bulbifera*. In the aqueous extract *S. asoca* contained the maximum amount of protein (1.27%) while the lowest (0.02%) was seen in *D. bulbifera*, *W. somnifera* and *A. indica* respectively. Overall maximum amount of total protein estimated in *A. sativum* extract and minimum amount in *D. bulbifera* extract in all extracts. Total sugars (%DM) results indicated that acetone extract of *P. guajava* contained the maximum i.e. 30.26% of total sugar, while the lowest was seen in aqueous extract of *A. barbadanis* (0.74%). Among the acetone extract *A. scholaris* (23.20%), *C. sativum* (22.58%) and *M. orvensis* (21.68%) also contain good quantity of sugars and *A. barbadanis* contained 2.28% minimum of total sugar. In the methanol extracts maximum total sugar was seen in *A. sativum* (28.00%) and minimum was seen in *S. mukorossi* (2.71%). Methanol extracts of *P. guajava* (20.18%) and *C. Sativum* (25.68%) are also rich in

sugar. The aqueous extract of *A. cepa* contained the maximum amount of total sugar (23.70%) and the second highest value was 22.09% in *A. sativum*. Highest amount of tannin was found in the methanol extract of *P. guajava* (22.28%), while the lowest tannin content was seen in the acetone extract of *T. belerica* (0.01%). Among the acetone extracts, *T. terrestris* contained the maximum amount of tannin (12.68%). High quantity of tannin was also found in *S. asoca* (12.00%) and in *A. sativum* (10.30%). Among the methanol extracts, *A. indica* contained the minimum tannin content (0.02%). In the aqueous extract, maximum tannin was obtained in *S. asoca* (9.02%), while the minimum was obtained in *A. indica* (0.06%). Among the different plant extracts, the methanol extract of *A. scholaris* contained the maximum saponin (29.32%), while the acetone extract of *A. barbadanis* contained the minimum amount of saponin (0.01%). The acetone extract of *A. cepa* contained the maximum amount of saponin (10.61%). The high amount of saponin content was

TABLE III
ANTIMICROBIAL ACTIVITY OF DIFFERENT PLANT EXTRACTS

S. No.	Local/ Common name	Botanical name	Acetone extract (mm)				Methanol extract(mm)				Aqueous extract(mm)			
			Strepto coccus	Staphy lococcus	Escherichia coli	Entero bacter	Strepto coccus	Staphy lococcus	Escherichia coli	Entero bacter	Strept ococcus	Staphy lococcus	Escherich ia coli	Entero bacter
1	Garlic	Allium sativum	NZ	NZ	6.5	NZ	NZ	NZ	6.0	NZ	NZ	NZ	6.5	NZ
2	Yam	Dioscorea bulbifera	NZ	NZ	NZ	7.5	NZ	NZ	NZ	8.0	NZ	NZ	NZ	NZ
3	Ashwgandha	withania somnifera	NZ	NZ	NZ	6.0	NZ	NZ	NZ	6.0	NZ	NZ	NZ	6.7
4	Arjuna	Terminalia arjuna	9.0	NZ	NZ	13.5	7.5	NZ	NZ	9.0	NZ	NZ	NZ	8.0
5	Guava	Psidium guajava	6.5	6.0	7.5	10.0	7.0	7.3	7.5	7.0	5.3	NZ	7.0	8.5
6	Alstonia	Alstonia scholaris	NZ	5.0	NZ	NZ	NZ	7.5	NZ	NZ	NZ	8.5	NZ	NZ
7	Eucalyptus	Eucalyptus globulus	8.0	9.0	NZ	12.0	7.0	8.0	NZ	12.5	NZ	7.0	NZ	11.0
8	Gokhru	Tribulus terrestris	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ
9	Reetha	Sapindus mukorossi	6.5	NZ	7.0	7.5	5.5	NZ	5.7	6.0	7.0	NZ	6.0	7.0
10	Ginger	Zingiber officinale	NZ	6.0	NZ	NZ	NZ	7.3	NZ	NZ	NZ	NZ	NZ	NZ
11	Behara	Terminalia belerica	12.5	NZ	NZ	16.0	10.0	NZ	NZ	12.0	NZ	NZ	NZ	7.0
12	Shikaki	Acacia concinna	5.3	7.0	6.5	10.0	NZ	7.0	6.5	8.0	NZ	7.0	6.0	6.5
13	Harad	Terminalia chebula	13.0	NZ	NZ	15.5	12.0	NZ	NZ	16.0	13.0	NZ	NZ	13.5
14	Jamun	Syzgium cumin	12.0	7.5	16.5	14.5	11.0	6.5	12.0	13.3	7.5	5.7	8.5	10.0
15	Neem	Azadiracta indica	8.5	7.0	5.0	9.5	9.0	7.0	5.0	4.7	NZ	7.0	5.0	5.0
16	Aloe vera	Aloe barbadanis	NZ	NZ	NZ	5.0	NZ	NZ	NZ	12.0	NZ	NZ	NZ	NZ
17	Satavari	Asparagus racemosus	NZ	7.0	NZ	NZ	NZ	7.7	NZ	NZ	NZ	8.3	NZ	6.0
18	Ashok	Saraca asoca	9.3	NZ	11.5	7.0	5.0	NZ	7.5	5.0	NZ	NZ	5.5	5.0
19	Coriender	Coriandrum sativum	NZ	NZ	5.0	5.0	NZ	NZ	6.5	5.0	NZ	NZ	5.0	NZ
20	Onion	Allium cepa	NZ	6.3	NZ	NZ	NZ	7.7	NZ	NZ	NZ	7.0	NZ	NZ
21	Pudina	Menthe orvensis	NZ	7.5	NZ	NZ	NZ	7.0	NZ	NZ	NZ	NZ	NZ	NZ
22	Std	Standard	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ	NZ

NZ- No Zone

also found in *C. sativum* (6.35%), *A. cepa* (6.29%) and *M. orvensis* (8.56%). In the methanol extract, saponin content was also high in *A. sativum* (20.94%), *P. guajava* (14.66%) and in *T. belerica* (15.32%). Minimum saponin content in this group was seen in *A. barbadanis* (0.19%). In the aqueous extract *A. scholaris* contained the maximum saponin (25.07%), while *A. barbadanis* having the minimum (0.06%).

The results of the effect of 21 plant extracts on in vitro methane production (ml / g DM, mM/g DM and mM/gDDM) and in vitro dry matter digestibility (IVDMD) were shown in table II. Among the different extracts tested the acetone extract of *E. globulus* showed the minimum methane production (0.21 ml/gm DM); there was 99.44% methane

reduction than the control. However, in most of the acetone extracts shown good methane reduction percentage. Out of 21 acetone extracts, in 7 plant extracts methane reduction was more than 60%. Methane reduction was 86.1, 64.45, 78.06, 99.44, 66.36, 75.76 and 60.91 percentage in *A. sativum*, *D. bulbifera*, *T. arjuna*, *E. globulus*, *S. mukorossi*, *T. belerica* and in *T. chebula*, respectively. But methane was not reduced by *A. racemosus* and *A. cepa* extracts. In acetone extracted group, highest increase in IVDMD (17.79%) was found in *Z. officinale*. The IVDMD percentage also increased in *A. barbadanis* (15.42 %), *C. sativum* (15.81 %) and *A. cepa* (16.47 %). Among the methanol extract the minimum methane production was observed in *E. globules* (22.81 ml/g DM) and

in *M. orvensis* (22.81 ml/g DM), where methane reduction was 39.74 % in both case. Results of other methanol extracts had also showed good percentage of methane reduction i.e. *A. sativum* (16.72 %), *T. chebula* (15.32 %) and *S. asoca* (16.72 %) respectively. IVDMD increased (7.80 %) only in *Z. officinale* among all methanol extracts but in most cases decrease in IVDMD percentage was found. Among the aqueous extracts *S. mukorossi* showed the minimum methane production (24.18 ml/g DM) and the methane reduction in terms of percentage was 17.89 %. Significant methane reduction in other extracts was also found in *P. guajava* (15.04%), *E. globulus* (16.26%) and in *T. chebula* (13.96%). Highest increase in IVDMD percentage was in *S. cumin* (7.36%) in comparison to other aqueous extracts. Methanol extracts of *E. globulus* and aqueous extract of *S. mukorossi* and *E. globulus* were emerged out the best inhibitor of methane.

The result of antimicrobial potential of plant extracts is shown in Table III. Among the various extract tested all the three extracts of *S. cumin* gave the best results with highest zone of 16.5mm against *Escherichia coli* (acetone extract). *T. terrestris* gave no zone against any bacteria tested. The *Enterobacter* was the most sensitive bacteria which were inhibited by 68% of the extracts tested. *E. globulus* gave no zone against *E. coli*, while *T. chebula* gave the best zone against *Enterobacter*. *P. guajava*, *A. indica*, *S. mukorossi*, *A. coneinna*, etc were some of the best inhibitors of bacterial growth which gave good zone against maximum culture tested.

IV. DISCUSSION

The potential of plant secondary metabolites to reduce enteric CH₄ production has been recognized and very extensive screening of a large range of plants and their secondary compounds, such as saponins and tannins, is now underway in several laboratories [36], [3]. Antimicrobial and other rumen fermentation effects induced by each plant species will depend on the nature, activity and concentration of its active principles. Differences in the activity of the secondary compounds of the plants are not only due to their different chemical nature, but also to different factors that can influence the concentration and activity of secondary metabolites within a given plant species. Some of the influential factors are origin, botanical variety, conditions of cultivation and harvesting, climatic and atmospheric factors, topographic factors, part of the plant used, processing and handling of the product like cleaning, drying, stabilization, conservation, storage, packaging, etc. [12]. In the present experiment, the highest methane reduction was found in acetone extract of *E. globulus* (99.44%). *E. globulus* which is also rich source of total tannin and total saponin i.e. 6.81% of DM and 1.08% of DM, respectively. Some plant extracts like *A. sativum* and *T. chebula*, which results in high methane reduction also contained of high amounts of total tannin and saponin. But, there are some plant extracts like *D. bublifera*, *T. arjuna*, *S. mukorossi* and *T. belerica*, which results in high methane reduction but contained less amount of total tannin and total saponin content. Our results are in also showed CH₄ suppressing effect with plants rich in saponins and the reason given related to their anti-protozoal effects[17]. Several

recent in vivo and in vitro studies provide evidence for CH₄ suppression for at least some saponin sources, although it seems that not all saponin sources are effective. The ability of condensed tannins to suppress CH₄ were given by [34], who tested in vitro the rumen fluid from sheep fed different CT plants. More recently, the diet of sheep supplemented with *Acacia mearnsii* CT and reduced CH₄ emissions by approximately 12% [23]. Evidently, some sources of CT are not effective in reducing CH₄ production, as shown for *Schinopsis quebracho* CT in cattle by [25]. In our study all the three extracts of garlic (*A. sativum*) is very good source of total tannin and saponin content and methane reduction potential was also very good in their acetone and methanol extract except aqueous extracts. They also possess antimicrobial activity against *Streptococcus*, *Staphylococcus* and *Anterobacter*. Our results were also in accordance with [38]. They reported that Allicin present in garlic is the main component responsible for the anti-bacterial and anti-fungal activity [15]. In our experiment with all three type of extracts of 21 different plant parts (i.e. 63 plant extracts), 34 plant extracts showed the methane reduction potential.

Methane reduction potential of various plant secondary compounds was also reported by various other workers. For example *Sesbania sesban*, *Acacia angustissima* [1], *Sapindus* sp., *Terminalia chebula*, *Populus tremuloides*, *Syzygium aromaticum*, *Psidium guayaba* [14], and with extracts of fennel, clove and garlic [4]. Some of the chemical compounds identified to be responsible for this effect were thymol, eugenol, carvacrol [22], tannins [18] or tea saponins [24]. Now, it is well established that direct or indirect inhibition of methane production entails a change in VFA profile which may increase the propionic acid or acetic acid production and more hydrogen was shifted for these acid production [13], [10]. Recently, it has been observed that the tannins could mitigate methane production in the rumen [18] and, in some cases, tannins found to reduce the degradation of the plant cell wall also [11], [7].

The saponins, tannins and different other bioactive compounds have been shown to affect the bacterial population in the rumen. Saponins from different sources have been found to be detrimental to protozoa and have been considered as possible defaunating agent in the rumen [37], [9]. Saponins react with the protozoal cell membrane leading to cell lyses and death. The steroidal saponin from *Yucca schidigera* alters the cell walls of the steroidal saponin inhibited non-cellulolytic bacteria. Steroidal saponins from *Y. schidigera* inhibit cellulolytic ruminal bacteria and fungi, but their effects on amylolytic bacteria were species dependent and similar to the effects of ionophores [40]. Other worker [39] reported that saponin influence both ruminal bacterial species and number through specific inhibition or selective enhancement of growth of individual species. Species dependent effects of saponin may affect bacteria with gram positive ultra structure than it does with those that are gram negative [35].

V. CONCLUSION

It was concluded from results that both acetone and methanol extracts of *E. globulus* and aqueous extract of *S. mukorossi* and *E. globulus* were the best inhibitor of methane production. This may be possibly due to high amount of

tannins or other bioactive compounds present in the extract. The *Enterobacter* was the most sensitive bacteria which showed the zone against 70 % of the extracts tested. The acetone extract was found to be more effective than methanol or aqueous extracts in inhibiting bacterial growth. Among all the extracts tested Jamun (*Syzygium cumini*) gave the maximum zone of inhibition for both Gram positive and Gram negative cultures.

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